

Acoustic Envelope

Turn supplied samples into one bounded event window

Calibrate quiet, retain a peak, publish relative evidence

Lesson	038 — component	Time	120–180 minutes
Prerequisites	Lessons 007, 012, and 037	Board	Mega 2560
Evidence	trace fixture and three LEDs	Status	host verified; bench open
Energy class	E0 copied observations; provisional E1 reference fixture		

Scope gate. This lesson interprets supplied **ENVELOPE** and optional **GATE** samples. It records no waveform, audio, or speech and makes no claim about sound-pressure level (SPL), loudness, frequency, source identity, hearing, impact severity, alarms, or surveillance. The optional physical route is only the SparkFun SEN-12642 with visible V10 silkscreen matched to the pinned hardware V1.0 sources.

1 Mission and vocabulary

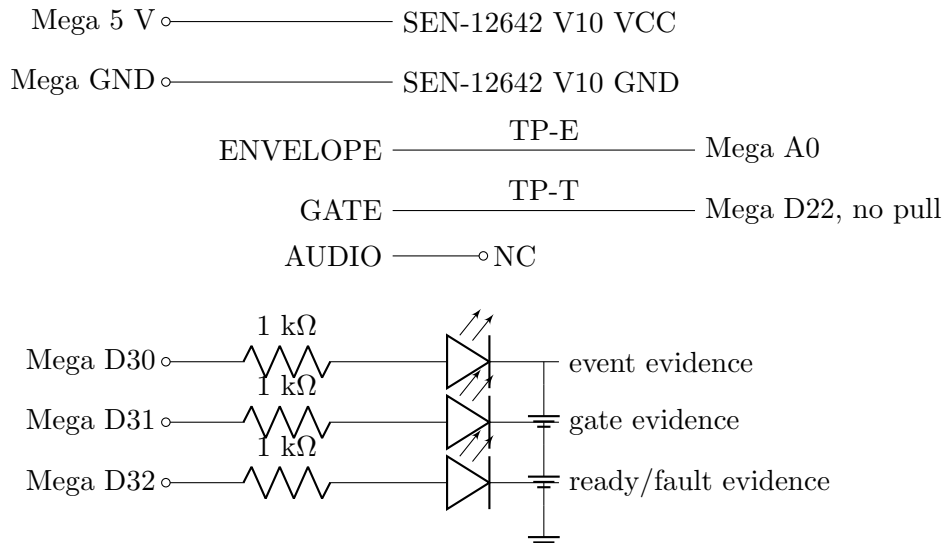
The evidence chain is:

supplied ADC/threshold trace → quiet baseline → event window → unitless intensity.

AcousticEnvelope is a pure behavior. It owns no endpoint, pin, interrupt, sampling timer, ADC reference, heap, callback, or hidden clock. The caller copies one ADC sample and, when configured, one threshold sample into each timestamped frame. The canonical sketch defaults to a deterministic trace, so the complete software experiment requires no acoustic hardware.

2 Read the exact reference circuit

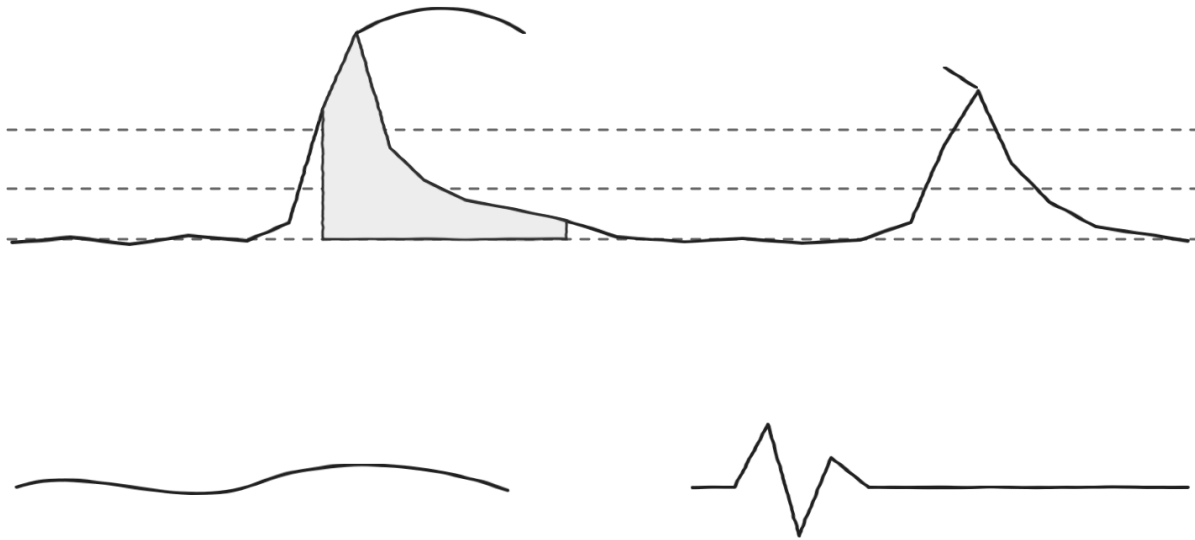
The optional reference detector, Mega, and USB source share one 5 V powered and unpowered state. Connect **ENVELOPE** to A0 at **TP-E**; connect active-high **GATE** to D22 at **TP-T** with no input pull-up. Leave **AUDIO** disconnected. The Mega ADC uses AVcc as its reference. D30, D31, and D32 each drive one LED through 1 kΩ to GND.



Electrically authoritative connection schematic for the named V10 reference fixture. D30, D31, and D32 are three independent branches; no LED, resistor, or node is shared.

This circuit does not authorize a look-alike sound board, KY-037/KY-038, or an unknown Elegoo module. Before power, match the visible V10 revision, trace the common ground, confirm AUDIO is open, and inspect for damage. The detector must never remain powered while the Mega is unpowered.

3 Read the supplied trace



Alternative text: a pencil ADC trace settles around a quiet baseline, crosses a configured attack offset, retains one peak, and closes after continuous quiet at a smaller release offset; a separate sparse trace shows that activity between supplied sample times may be missed.

Dashed lines, from low to high, mark baseline, release, and attack. Vertical strokes mark event start and quiet close; the shaded region is the bounded event window. The lower strokes contrast slow quiet adaptation with a narrow excursion between supplied samples.

The first healthy, non-clipped sample establishes the baseline and starts calibration with zero amplitude. During calibration and quiet, the integer estimate follows:

$$B_{\text{next}} = B + \text{trunc}\left(\frac{\text{raw} - B}{2^{\text{baselineShift}}}\right).$$

It freezes while an event or refractory interval is active. Each amplitude is $|\text{raw} - B|$. Attack requires amplitude at least the configured attack offset; release requires amplitude at most the smaller release offset. That gap prevents one noisy boundary from repeatedly opening and closing.

Calibration includes the first sample at or beyond its configured duration. Before that boundary, no event is qualified. A rail sample before calibration starts does not establish a baseline; clipping during calibration faults and requires reset.

4 Open, retain, and close one event

An attack opens one bounded window and freezes the baseline. The window retains the greatest observed amplitude. It closes at the earlier of continuous quiet for `quietToClose` or exact `eventWindow` expiry. Refractory time starts at completion and blocks another event until its exact boundary.

Only the completion update publishes relative intensity. Define:

$$H = \max(B - \text{railMargin}, (1023 - \text{railMargin}) - B).$$

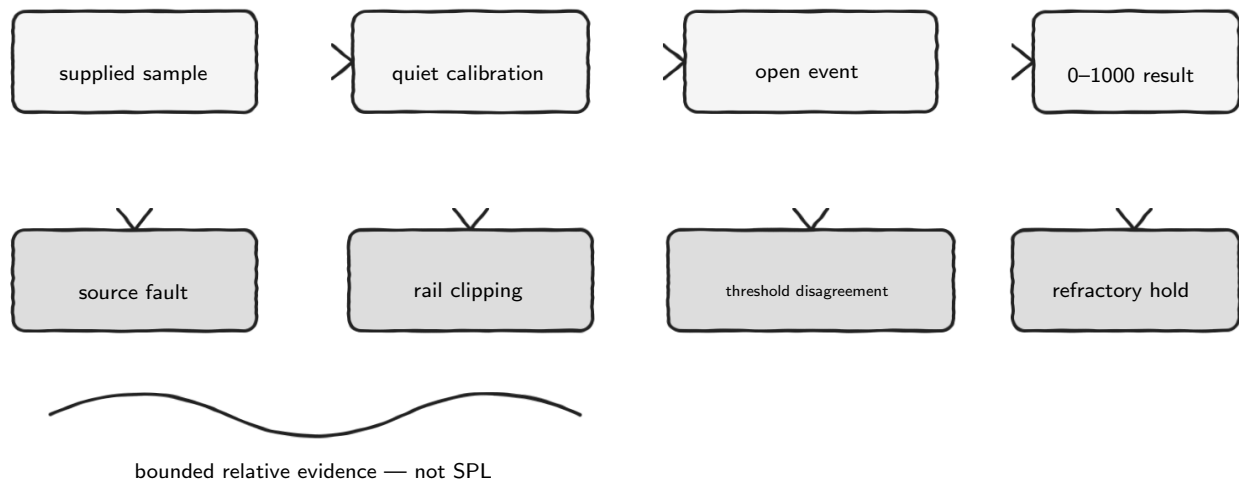
For peak P , attack offset A , and valid $H > A$:

$$I = \frac{1000 \cdot \text{clamp}(P - A, 0, H - A)}{H - A}.$$

Integer division truncates. I is exact replay evidence in the range 0–1000 for this configuration and baseline. It is unitless: do not compare it to dB SPL, another board, another gain setting, another baseline, or another room as though those values share a physical scale.

Published field	Interpretation
<code>raw</code> , <code>baseline</code> , <code>amplitude</code>	current supplied sample and derived offset
<code>peakAmplitude</code>	largest supplied amplitude in the open window
<code>eventStarted/Completed</code>	one-update transition evidence
<code>eventStartedAt/Duration</code>	bounded interval over supplied timestamps
<code>relativeIntensity</code>	completed-event value, 0–1000, unitless
<code>phase/quality/status</code>	whether any of the above is qualified

5 Confidence before interpretation



Alternative text: a pencil flow runs from a supplied ADC and optional threshold sample through quiet calibration and an event window to a unitless 0–1000 result; side branches stop for source failure, rail clipping, sustained threshold disagreement, and refractory time.

Top row, left to right: supplied sample, quiet calibration, open event, unitless result. Downward branches, left to right: source fault, rail clipping, sustained threshold disagreement, and refractory hold.

`raw <= railMargin` is `ClippedLow`; `raw >= 1023-railMargin` is `ClippedHigh`. Clipping cannot open or extend an event. A failed analog source is not interpreted at all. A configured threshold source must also be healthy.

After calibration, the optional digital threshold is compared with the analog authority: active should agree with amplitude at or above attack. One mismatch is visible evidence, not a fault. Continuous mismatch for `quietToClose` becomes `ThresholdDisagreement`. Comparator output never replaces the analog envelope for timing or intensity, and a threshold-only specimen is unsupported.

Phase	Quality	Status	Event-field rule
Calibrating	Unqualified	Ok	event fields canonical zero
Quiet	ValidQuiet	Ok	event fields canonical zero
EventOpen	ValidEvent	Ok	live start/peak; completion and intensity false/zero
Refractory	ValidEvent	Ok	completion update publishes nonzero duration
Refractory	ValidQuiet	Ok	completion snapshot cleared; no new attack
Fault	ClippedLow/High	InvalidArgument	event and intensity fields cleared
Fault	ThresholdDisagreement	HardwareFailure	event and intensity fields cleared
Fault	SourceFault	exact source status	event and intensity fields cleared
Fault	SourceFault	HardwareFailure	configured threshold presence mismatch; event/intensity cleared
Fault	SourceFault	InvalidArgument	malformed runtime threshold enum; event/intensity cleared
Fault	TimingFault	InvalidArgument	last completed evidence retained
Fault	Unqualified	InvalidArgument	raw above 1023 or invalid headroom; event/intensity cleared

No other phase, quality, and status tuple is legal. “Exact source status” means the non-Ok analog status, or the configured threshold status after a healthy analog sample. A malformed runtime threshold `Level` is instead `Fault/SourceFault/InvalidArgument`. A supplied raw value above 1023 and invalid runtime headroom are both specifically `Fault/Unqualified/InvalidArgument`; unlike a timing fault, it clears event and intensity fields and requires reset and recalibration.

Invalid construction policy is different from a runtime fault: `initialize()` returns `InvalidArgument`, leaves the behavior uninitialized, and does not mutate the snapshot into `Fault`.

6 Transition order and canonical evidence

After the initialized-state check, each update applies this validation precedence: timestamp validity; malformed configured runtime threshold enum; analog status; configured-threshold presence agreement; configured threshold status; raw range 0..1023. Only then does interpretation apply clipping, calibration, sustained disagreement, event close, event open, and baseline update. A winning fault prevents every lower transition.

Before completion, relative intensity, event start, and event duration use canonical zeros. Clipping, source failure, and disagreement also publish zero event/intensity fields. A timing fault retains the last completed evidence but still reports `Fault/TimingFault/InvalidArgument`; status and quality must always travel with copied values.

An identical same-time frame is idempotent. Changed same-time evidence, an apparent backward timestamp, or a jump at least the unsigned half-range faults without partial mutation. Natural 32-bit rollover remains valid.

7 Narrative trace fixture

```
void loop ()
{
    observeSuppliedTrace ();

    if (!classifyAcousticWindow ())
    {
        presentFaultEvidence ();
        return;
    }

    presentRelativeEvidence ();
}
```

The fixture begins with midrail quiet samples, crosses attack, retains a peak, returns below release long enough to close, then waits through refractory. Separate traces exercise positive and negative excursions of equal amplitude, slow baseline drift, clipping, source failure, threshold chatter and sustained disagreement. Replaying identical frames must produce byte-identical snapshots. The deterministic fixture advances evidence time by 20 ms per frame. Physical reference mode instead timestamps acquired samples with real elapsed `millis()`.

8 Predict, observe, interpret

1. Predict the baseline from the first healthy midrail sample. Observe calibration through its exact boundary; interpret no event before it.
2. Supply slow quiet drift. Predict the integer baseline steps, including truncation toward zero.
3. Supply attack-offset minus one, exact attack, and plus one. Predict only the exact and greater samples can open a window.
4. Supply equal positive and negative excursions. Predict equal amplitude; retain the greater peak when several samples share one window.
5. Close by continuous quiet, then by maximum-window expiry. Predict the earlier condition wins and intensity appears only on completion.
6. Supply rail-margin minus one, exact low clipping, and exact high clipping. Predict a fault, zero event evidence, and reset requirement.
7. With an optional threshold trace, compare one mismatch with sustained disagreement. Interpret the analog envelope as authority in both cases.
8. Leave a narrow pulse entirely between timestamps. Predict that the supplied trace contains no evidence of it; do not claim complete capture.

9 Sampling, privacy, and stop rules

This is a polled observation model. Its documented update interval is part of the evidence. A pulse wholly between updates may be missed; supplied points cannot establish bandwidth, frequency response, alias rejection, onset between samples, or continuous capture. The component stores only bounded state and completed relative evidence, never a waveform or audio recording.

Stop rule. Use the default host trace unless the detector is the exact V10 reference and the unpowered checks pass. Never ask for a loud or startling stimulus. Stop for an unidentified part, out-of-rail output, unexpected reset, rail instability, heat, odor, excessive current, or specimen/schematic disagreement. Do not use this lesson to record speech, monitor people, infer medical or safety conditions, or claim SPL.

10 Diagnosis, exercises, and evidence

Observation	Interpret and inspect
baseline never qualifies window chatters	startup clipping, source status, calibration time strict attack/release relationship and supplied times- tamps
peak changes in refractory threshold disagrees once	reject the trace implementation; baseline/peak freeze retain mismatch evidence; sustained duration not met
same trace yields different bytes	hidden time/state or noncanonical fields
similar tap yields different intensity	baseline, headroom, gain, distance, room; not SPL

1. Calculate three baseline steps for a chosen shift and asymmetric noise.
2. Compute intensity for attack, an interior peak, and a clamped peak.
3. Explain why headroom depends on baseline and rail margin.
4. Order clipping, disagreement, event completion, and event opening when several conditions occur in one frame.
5. State exactly what cannot be inferred from a 0–1000 value.

Implemented focused host evidence covers every configuration boundary; calibration and timing edges; baseline arithmetic; positive/negative peaks; intensity endpoints; quiet and maximum-window close; refractory; rails; source faults; optional threshold agreement; same-time and rollover cases; lifecycle; deterministic replay. The canonical Mega sketch defaults to the same hardware-independent trace and presents event completion, copied gate state, and readiness through three independent LEDs. Setting its reference-fixture switch reads `ENVELOPE/A0` and active-high `GATE/D22`.

D30 and D31 are commanded off before the first event, separately proving the safe inactive output intent. D32 turns on only after the resources required by the selected path and the envelope behavior initialize successfully. Physical mode owns and acquires A0 through `AnalogInput` and D22 through a no-pull `DigitalInput` before any indicator. The absent-hardware trace acquires no physical source, so D32 makes no physical-acquisition claim. Shutdown releases outputs first, then physical inputs, then resets behavior. After stop, measure D30, D31, and D32 at their pin-side test points for inactive/high-impedance state. Host tests, not LED darkness, establish that resources were released.

Compilation, this PDF, and host traces are not microphone, ADC bandwidth, privacy, acoustic, SPL, or physical acceptance evidence. Incoming revision matching, measured output ranges, safe-state evidence, and the complete E1 card remain open. The HTML lesson is the accessible alternative; this print-edition PDF remains untagged and makes no PDF/UA claim.

11 Reference design sources

- SparkFun Sound Detector product page (sparkfun.com).
- SparkFun Sound Detector hookup guide (learn.sparkfun.com).
- SparkFun Sound Detector hardware V1.0 source, pinned at commit: `a456d9e5e916aa3cfe656a48f5670cb73323f696`.
- Texas Instruments LMV324 low-voltage operational-amplifier datasheet (ti.com).